ISAIAH ALFRED MARTINEZ

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EDUCATION

University of Nebraska - Lincoln — Lincoln, NE
Ph.D. in Mathematics — *Aug. 2024 – Present*
Advisor: Mark Walker

California State University, Fresno — Fresno, CA
M.S. in Mathematics — *Aug. 2022 – May 2024*
Thesis: Khovanov homology for singular links and tangles
Advisor: Carmen Caprau

University of California, Santa Barbara — Santa Barbara, CA
B.S. in Mathematics — *Sept. 2018 – Mar. 2022*

RESEARCH

My current research interests lie in the application of homological methods to commutative algebra. Previously, my research interests were more topological in nature.

PUBLICATIONS

TRI-PLANE DIAGRAMS FOR SIMPLE SURFACES IN S^4 — *Jun. 2023*
(joint w/ many authors) Journal of Knot Theory and its Ramifications, Vol. 32, No. 6

TALKS AND POSTERS

To see selected slides/notes/posters, see my website.

Constructing non-Koszul quadratic Gorenstein algebras II — *Oct. 2026*
RTG Seminar on Koszul Algebras — *Lincoln, NE*

Constructing non-Koszul quadratic Gorenstein algebras I — *Oct. 2026*
RTG Seminar on Koszul Algebras — *Lincoln, NE*

The Singularity Category of a Gorenstein Ring — *Sept. 2026*
Commutative Algebra Reading Seminar (CARS) @ UNL — *Lincoln, NE*

Transfer of Koszulness — *Apr. 2026*
RTG Seminar on Koszul Algebras — *Lincoln, NE*

Ĉech vs De Rham cohomology — *Dec. 2025*
Cohomology Club @ UNL — *Lincoln, NE*

The tensor product of Hilbert spaces is (almost) a Hilbert space — *Nov. 2025*
(joint w/ Audrey Baumheckel) Operator Theory Reading Seminar (OTRS) @ UNL — *Lincoln, NE*

Injective resolutions and Bass numbers — *Sept. 2025*
Commutative Algebra Reading Seminar (CARS) @ UNL — *Lincoln, NE*

CW and cellular cohomology — *Aug. 2025*
(joint w/ Anna Brosowsky) Cohomology Club @ UNL — *Lincoln, NE*

An introduction to Khovanov homology — *Nov. 2024*
Groups-Semigroups-Topology (GST) Seminar @ UNL — *Lincoln, NE*

Khovanov homology for singular links and tangles *May 2024*
M.S. Thesis Defense *Fresno, CA*

Khovanov homology for singular links and tangles *May 2024*
BAMM!Fest 2024 *San Francisco, CA*

On an invariant for singular knots *Apr. 2024*
45th Annual Central California Research Symposium *Fresno, CA*

An extension of Khovanov homology to singular links *Jan. 2024*
JMM 2024 *San Francisco, CA*

A crash course in Khovanov homology *Nov. 2023*
Department of Mathematics Day, Fresno State *Fresno, CA*

An extension of Khovanov homology to singular links *Nov. 2023*
2nd CSU Mathematical Sciences Research Conference *Bakersfield, CA*

Tri-plane diagrams and surface knot invariants *May. 2023*
BAMM!Fest 2022 *Fresno, CA*

Tri-plane diagrams and surface knot invariants *Apr. 2023*
44th Annual Central California Research Symposium *Fresno, CA*

Tri-plane diagrams for simple surface knots *Jan. 2023*
(joint w/ Wolfgang Allred, Zack Dooley, & Alec Goldman) JMM 2023 *Boston, MA*

Category theory and homological algebra *May 2022*
(joint w/ Joseph Sullivan) UCSB DRP Poster Session *Santa Barbara, CA*

Quadratic forms, the Witt ring, and Frobenius reciprocity *May 2020*
UCSB DRP Virtual Poster Symposium *Santa Barbara, CA*

Teaching Experience

University of Nebraska — Lincoln *Aug. 2024 – Present*

Associate Convener

Responsibilities: In addition to the teaching responsibilities as an instructor of record, I worked closely with the faculty convener to prepare for the semester, run weekly course meetings, and generally support the other course instructors on a day-to-day basis. Additional responsibilities include but are not limited to: making course schedule, scheduling WeBWork homework, observe and give feedback to graduate student instructors, help with graduate student orientation.

- Math 100A: Intermediate Algebra — Fall 2026, Spring 2027

Instructor of Record

Responsibilities: Full responsibility for course activities including providing weekly lectures and facilitating active learning. Responsible for grading and submitting student grades to the university at the end of each semester. Responsible for holding weekly office hours.

- Math 100A: Intermediate Algebra — Fall 2025

Teaching Assistant

Responsibilities: Responsible for facilitating active learning activities which reinforce the concepts students have seen in lecture in the previous weeks. Responsible for grading student work often including exams and holding weekly office hours.

- Math 107: Calculus II — Fall 2024 (2 sections), Spring 2025 (2 sections), Summer 2025

- Math 208: Calculus III — Spring 2026 (2 sections)

Grader

Responsibilities: Responsibilities vary depending on course needs each semester. Usually this includes grading some subset of homework assignments, quizzes, and/or exams.

- Math 104: Applied Calculus — Spring 2026

California State University, Fresno

Teaching Associate *Aug. 2022 – May 2024*

Responsibilities: Responsible for teaching three "lab" courses each semester through a combination of lecture and active learning activities. Responsible for evaluating student assignments and maintaining/submitting student records (grades). Responsible for holding office hours every week.

- Math 3L: College Algebra — Fall 2023
- Math 10AL: Structure & Concepts in Mathematics I — Spring 2024
- Math 11L: Elementary Statistics — Fall 2022, Spring 2023, Fall 2023 (2 classes)
- Math 45L: What is Math? — Fall 2022 (2 classes), Spring 2023 (2 classes), Spring 2024 (2 classes)

University of California, Santa Barbara

Course Reader (Grader) *Mar. 2021 – Dec. 2021*

Responsibilities: Provided thoughtful and constructive feedback to homework and assigned grades to every homework assignment. Responsible for submitting grades to "GauchoSpace."

- Math 8: Transition to Higher Mathematics — Spring 2021, Fall 2021

MENTORING

UNL Math Grad Student Peer Mentor *Fall 2026*

- Each fall our graduate program assigns a senior grad student (on a volunteer basis) to mentor one of our incoming first year grad students. We help students assimilate to graduate student life in general and give advice including but not limited to: searching for a research area, searching for an advisor, how to study for qualifying exams, finding work/life balance, etc.
- Fall 2026: Mentee — Jenna Shuffelton

UNL First Year First Generation REU Mentor *Summer 2026*

- Served as one of the primary mentors for four incoming UNL undergraduates.
- Organized weekly panel participants to give advice and share first-gen experiences regarding university life.
- Helped organize professional development workshops including LaTeX/beamer, coding, and CV writing.
- Helped introduce students to basic set and graph theory so that they could work on a graph theoretical research question that has commutative algebra implications.

UNL Directed Reading Program Mentor

- Fall 2026: Mentee — Lewis Rokke
- Fall 2025: Mentee — Eric Woods

- Spring 2025: Mentee — Noah Crawford

UCSB Math Major Peer Mentor with Mentor Collective *Nov. 2021 - June 2022*

- Certified trained mentor with Mentor Collective.
- Mentored four undergraduate STEM students in the 2021-2022 academic year, whose concerns ranged from academic performance to seeking employment upon graduation.

SERVICE

UNL Math Day Committee Member *Fall 2026 – Present*

- Helping with the running of Math Day at UNL on a more administrative level. I've helped with general logistics in committee meetings.
- Assisted with the writing of "Probe 1" (a math exam high school students take to qualify to participate in our quiz bowl).
- Assisted with the writing of Quiz Bowl problems.

Nebraska Conference for Undergraduate Wisdom in Mathematics (NCUWM) 2026 *Jan 2026*

- Gave campus tour to visiting undergraduate participants.

UNL Math Day *Nov. 2025*

- Proctored Quiz Bowl rounds.

Nebraska Conference for Undergraduate Women in Mathematics (NCUWM) 2025 *Jan. 2025*

- Helped organize folders with conference information and welcome bags handed out to participants.

Math Field Day 2023 (Fresno State) *Apr. 2023*

- Math field day is an outreach program held annually at Fresno State that attempts to foster mathematical curiosity in local middle school and high school students.
- Assisted with the proctoring of a mathematics contest for high school students.

Sonia Kovalevsky Day 2023 & 2024 *Mar. 2023/2024*

- Sonia Kovalevsky Day is an annual outreach event held at Fresno State that aims to inspire the next generation of female mathematicians, scientists, engineers, and innovators.
- Assisted with an assortment of mathematics activities and games in an effort to showcase the fun and creative ways one can think about mathematics.

CONFERENCES ATTENDED

KUMUNU 2026 *Oct. 2026*

URiCA 2026 (Upcoming Researchers in Commutative Algebra) *Aug. 2026*

CA+ 2026 *Mar. 2026*

KUMUNU 2025 *Oct. 2025*

Summer CAMP (Commutative Algebra Market Prepartation) @ UNL *Aug. 2025*

URiCA 2025 (Upcoming Researchers in Commutative Algebra) *May 2025*

RayFest (conference celebrating the work and retirement of Ray Heitmann) *Apr. 2025*

Commutative algebra in mixed characteristic @ UNL *Mar. 2025*

Joint Mathematics Meeting 2024 *Jan. 2024*

CSU Mathematical Sciences Conference 2023 *Nov. 2023*

Field of Dreams 2023 *Nov. 2023*

AMS Spring Western Sectional Meeting 2023 *May 2023*

44th Annual Central California Research Symposium *Apr. 2023*
Joint Mathematics Meeting 2023 *Jan. 2023*
Field of Dreams 2022 *Nov. 2022*
Math For All 2021 (hosted virtually by Tulane University) *Mar. 2021*
The California Forum for Diversity in Graduate Education *Oct. 2020*
Pacific Math Alliance Conference 2019 *Oct. 2019*

AFFILIATIONS

American Mathematical Society | *Graduate Student Member* *Sept. 2022 – Present*
Math Alliance | *F-GAP Scholar* *Apr. 2022 – May 2024*

SCHOLARSHIPS, HONORS, AND GRANTS

Fresno State College of Science and Mathematics Outstanding Thesis *2024*
Fresno State Department of Mathematics Outstanding Thesis *2024*
Graduate Student Ambassador for the Mathematics M.S. program at Fresno State *2022-2023*
Fresno State Division of Research and Graduate Studies Graduate Student Travel Grant *2022*
Bolstering the Advancement of Masters in Mathematics (BAMM!) Scholarship *2022–2024*
Edison GRE Scholarship *2020*
UCSB General Scholarship *2018 – 2021*
BLOSS MHS Alumni Scholarship *2018,2020,2021*

SOFTWARE PROFICIENCY

LaTeX, Python, Macaulay2, Github